A

Mini Project Report on

VIDEO ANALYSIS FOR REAL-TIME VEHICLE CLASSIFICATION AND COUNTING

Submitted for partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

by

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St. MARTIN'S ENGINEERING COLLEGE

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Certificate

This is to certify that the project entitled **"VIDEO ANALYSIS FOR REAL-WORLD VEHICLE CLASSIFICATION AND COUNTING"** is being submitted by S.BHAVITHA(22K81A05H8) Y.SWARNA(22K81A05K2) in fulfilment of the requirement for the award of degree of **BACHELOR OF TECHNOLOGY** in **COMPUTER SCIENCE AND ENGINEERING** is recorded of bonafide work carried out by them. The result embodied in this report have been verified and found satisfactory.

Signature of Guide	Signature of HOD
Mr. Praneel Deva	**Dr. R. Santhosh Kumar**
Assistant Professor	Professor and Head of Department
Department of CSE	Department of CSE

Internal Examiner **External Examiner**

Place:

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

We, the students of "**Bachelor of Technology in Department of Computer Science and Engineering"**, session: 2022 - 2026**, St. Martin's Engineering College, Dhulapally, Kompally, Secunderabad,** hereby declare that the work presented in this project work entitled **"VIDEO ANALYSIS FOR REAL-WORLD VEHICLE CLASSIFICATION AND COUNTING "** is the outcome of our own bonafide work and is correct to the best of our knowledge and this work has been undertaken taking care of Engineering Ethics. This result embodied in this project report has not been submitted in any university for award of any degree.

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ABSTRACT

- Real-time vehicle classification and counting play a crucial role in traffic management, urban planning, and intelligent transportation systems. Traditional traffic monitoring techniques rely on manual observation or basic image processing, which often lack accuracy and scalability. This project proposes an AI-powered Video Analysis System that utilizes deep learning techniques, including Convolutional Neural Networks (CNNs) and object detection models such as YOLOv5, to classify and count vehicles in real time. By integrating AI-driven image analysis and automated tracking, the system enhances efficiency in traffic monitoring, congestion management, and road safety applications.

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CHAPTER 1
INTRODUCTION

Accurate vehicle classification and counting are essential for modern traffic monitoring systems. Conventional approaches use motion sensors, manual counting, or rule-based image processing, which are often inefficient, expensive, and error-prone.

Deep learning and computer vision offer advanced solutions by leveraging real-time video analysis for object detection and classification. YOLOv5, a state-of-the-art object detection model, provides a robust framework for vehicle identification, ensuring fast and accurate results.

This project aims to develop an AI-powered system for real-time vehicle detection, classification, and counting. The system will process live traffic footage, detect different vehicle types (cars, buses, trucks, motorcycles), and generate real-time traffic insights to aid in traffic management and urban planning.

By adopting AI-driven video analysis, transportation authorities and city planners can optimize traffic flow, reduce congestion, and enhance road safety.

CHAPTER 2
LITERATURE SURVEY

Brown, T., & White, M. (2019). “Deep Learning for Traffic Monitoring Systems." This study explores CNN architectures for vehicle detection

Singh, R., & Patel, K. (2020)."AI-Based Vehicle Classification in Smart Cities." The research discusses AI-driven vehicle categorization techniques.

Nguyen, L., & Sharma, P. (2021). "YOLO-Based Vehicle Detection for Traffic Analytics." This paper evaluates YOLO models for real-time vehicle analysis.

Martinez, D., & Johnson, S. (2022). "Intelligent Transportation Systems Using Computer Vision." The authors present AI-driven traffic management solutions.

.**Garcia, H., & Lewis, F. (2023).). "Real-Time Vehicle Counting and Road Traffic Analysis."** This study examines AI techniques for improving traffic flow monitoring

Chen, W., & Roberts, L. (2023).). "Edge Computing for AI-Powered Traffic Surveillance." The paper explores AI-driven geospatial data processing.

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CHAPTER 3
SYSTEM ANALYSIS AND DESIGN

3.1 Existing System

Traditional traffic monitoring systems rely on:

- Manual observation, leading to human errors and inefficiencies.
- Inductive loop sensors, which require expensive installation and maintenance.
- Basic image processing methods that struggle with occlusions and varying weather conditions.
- Limited scalability for large-scale deployments.

3.2 Proposed System

The proposed AI-based vehicle classification and counting system integrates deep learning and computer vision to improve traffic monitoring accuracy and efficiency. Key advantages include:

- Real-time vehicle classification with high accuracy using YOLOv5.
- Automated vehicle counting and tracking for traffic flow analysis.
- Scalable architecture suitable for smart city applications.
- AI-driven insights for better traffic management and congestion control.

3.3 System Configuration

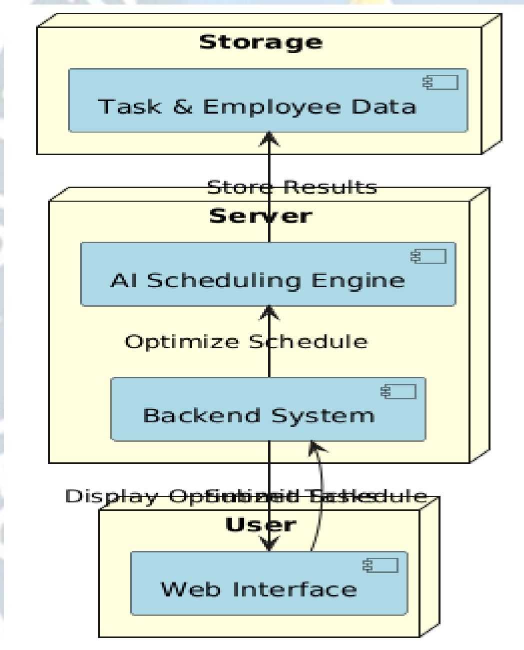
Software Requirements

- **Development Tools**: Python, OpenCV, TensorFlow, PyTorch.
- **AI Models**: YOLOv5, DeepSORT, Faster R-CNN.
- **Libraries**: NumPy, Pandas, Scikit-learn, Matplotlib.
- **Data Sources**: Public traffic datasets (UA-DETRAC, KITTI, COCO Traffic).

Chapter 4

4.1 Design:

4.1.1 System Architecture



4.1.2 Data Flow Diagram

4.1.3 UML Diagram

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Encourage the growth of OO tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns and components.

4.1.4 Use Case Diagram

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

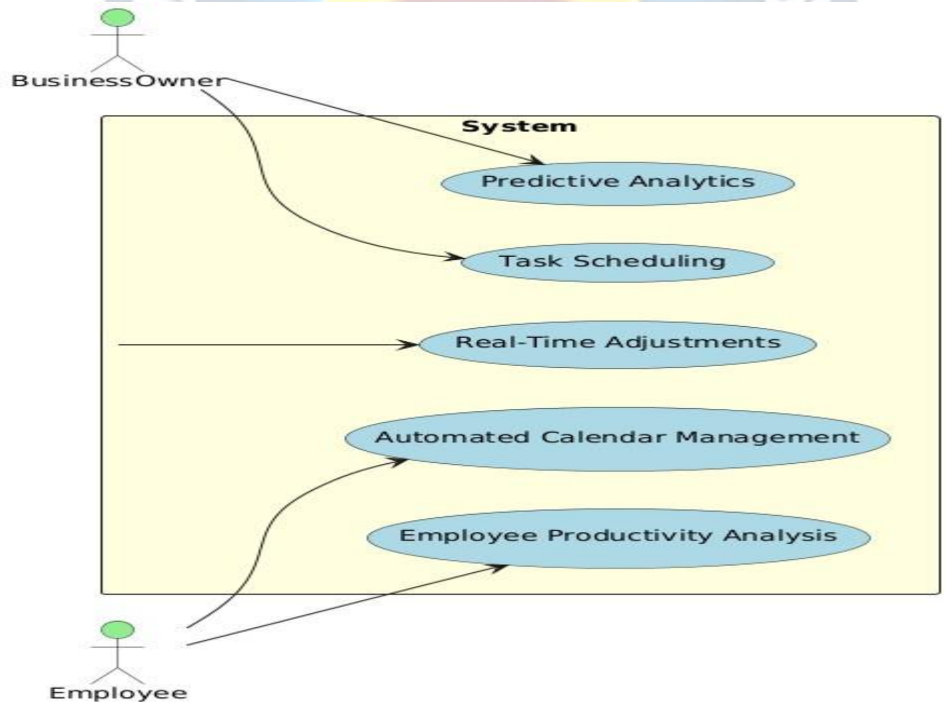


FIG: Use case Diagram

4.1.5 Activity Diagram

Activity diagrams are graphical representations of workflows of stepwise activities and actions[1] with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams are intended to model both computational and organizational processes (i.e., workflows), as well as the data flows intersecting with the related activities.[2][3] Although activity diagrams primarily show the overall flow of control, they can also include elements showing the flow of data between activities through one or more data stores.[citation needed]Activity diagrams are graphical representations of workflows of stepwise activities and actions[1] with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams are intended to model both computational and organizational processes (i.e., workflows), as well as the data flows intersecting with the related activities.[2][3] Although activity diagrams primarily show the overall flow of control, they can also include elements showing the flow of data between activities through one or more data stores.[citation needed]

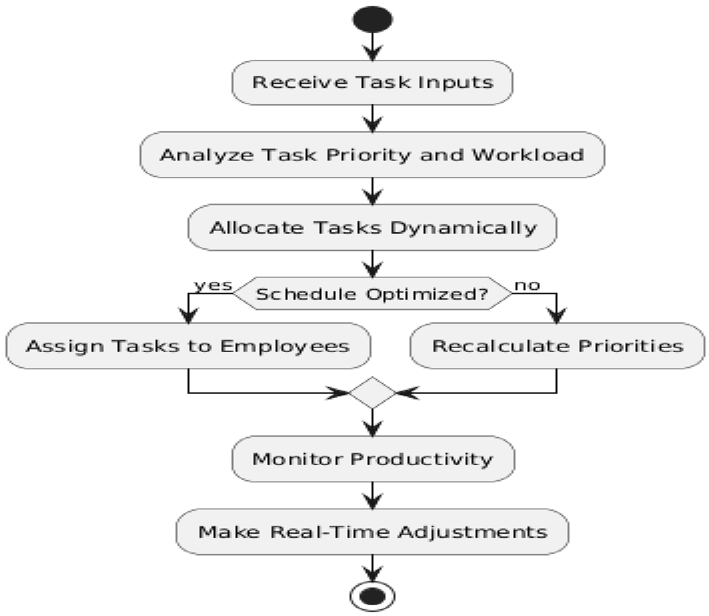


FIG: Activity Diagram

4.1.6 Sequence Diagram

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams

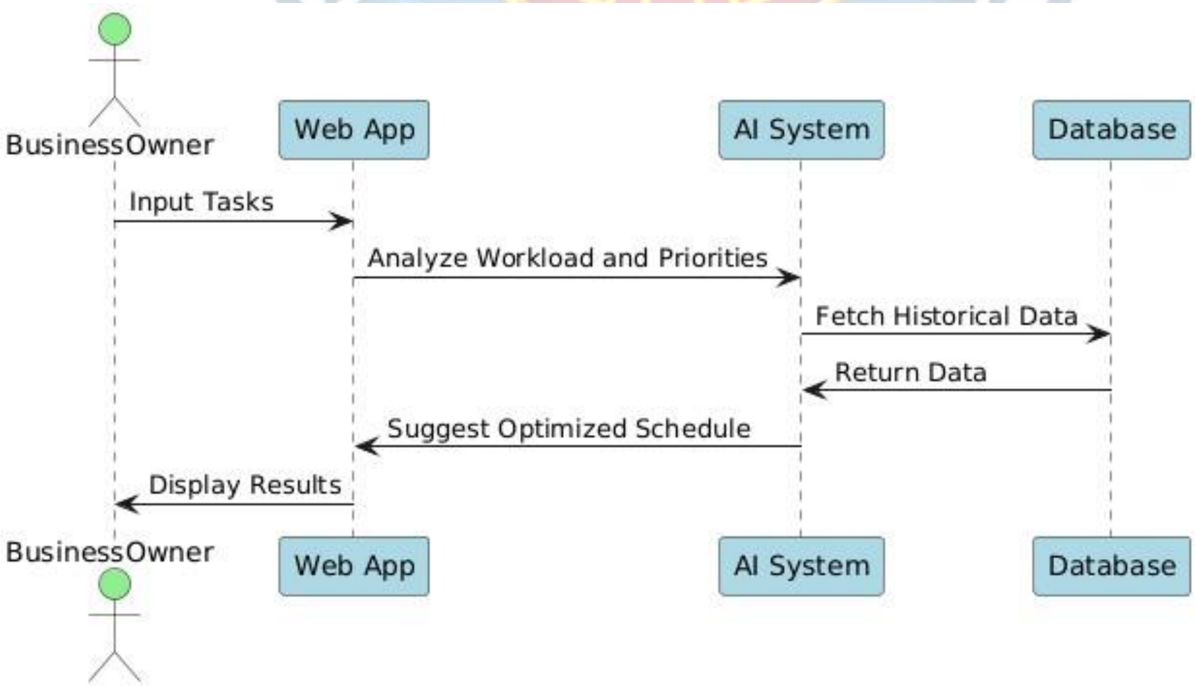


FIG: Sequence diagram

4.1.7 Class Diagram

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes,

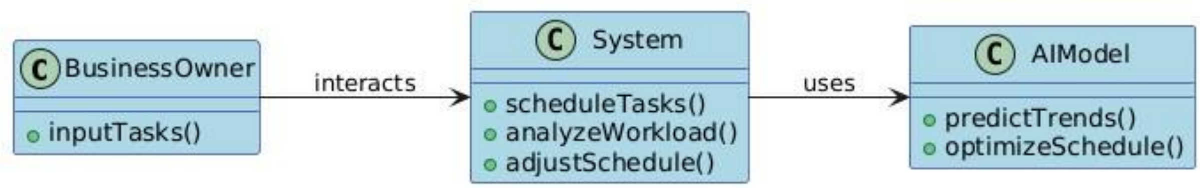


Fig : class diagram

Modules

Image Acquisition and Preprocessing Module: Collects and normalizes noisy images for analysis.

Deep Autoencoder-Based Denoising Module: Uses CAEs and VAEs for learning noise patterns and reconstructing clean images.

Feature Retention and Enhancement Module: Ensures important image details are preserved.

Performance Evaluation Module: Compares the denoised images with ground truth data for accuracy assessment.

Visualization and Reporting Module: Provides analytics and monitoring of denoising performance.

4.5 System Requirements

4.5.1 Software Requirements

- **Development Tools**: Python, OpenCV, TensorFlow, PyTorch.
- **AI Models**: YOLOv5, DeepSORT, Faster R-CNN.
- **Libraries**: NumPy, Pandas, Scikit-learn, Matplotlib.
- **Data Sources**: Public traffic datasets (UA-DETRAC, KITTI, COCO Traffic).

CHAPTER 5
SOURCE CODE

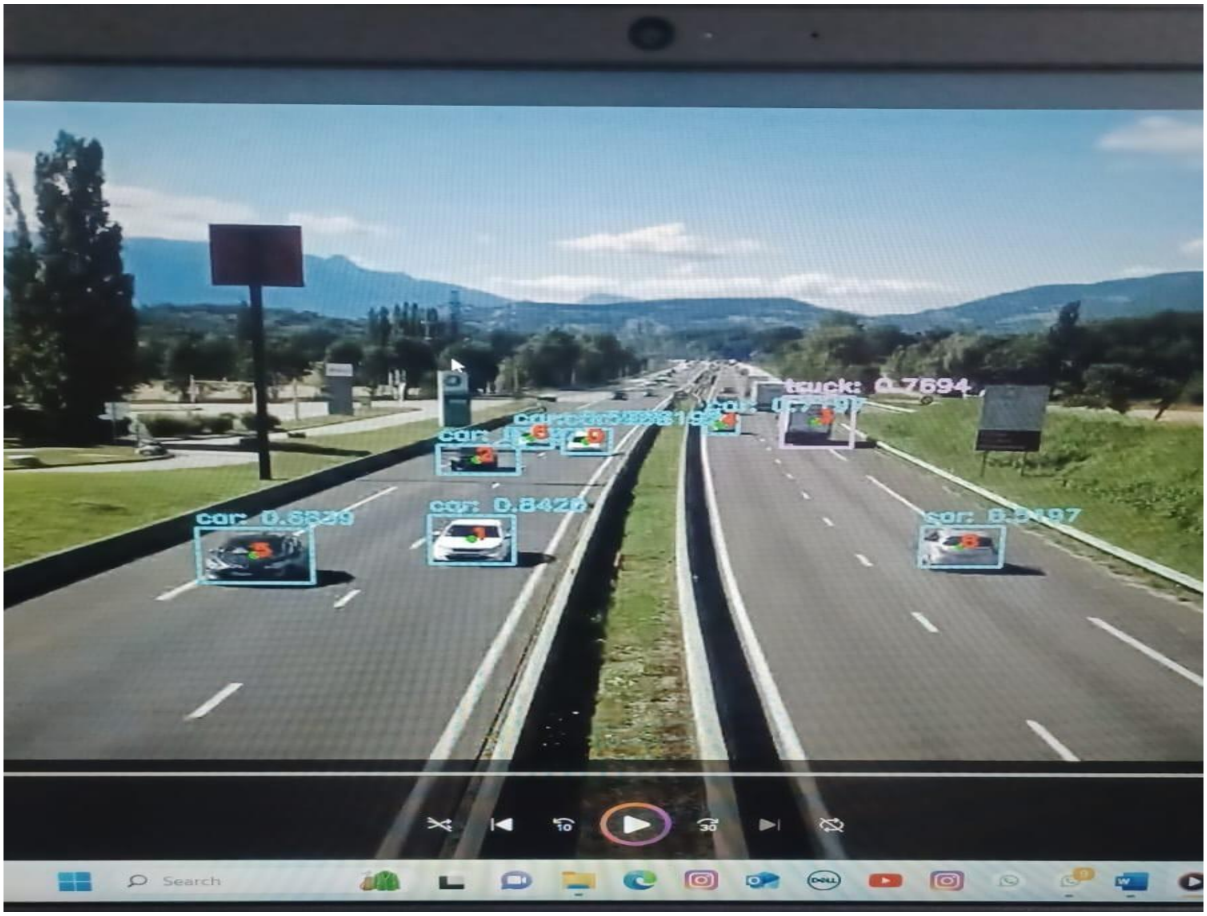
```
import tkinter as tk
from tkinter import filedialog
from tkinter import *
import os
import subprocess
import numpy
#initialise GUI
top=tk.Tk()
top.geometry('1200x720')
top.title('RealTime Vechicle Project')
bg = PhotoImage(file = "a.png")
canvas1 = Canvas( top, width = 800, height = 800)
canvas1.pack(fill = "both", expand = True)
# Display image
canvas1.create_image( 0, 0, image = bg, anchor = "nw")
# top.configure(background= bg)
label=Label(top,background='#CDCDCD', font=('arial',15,'bold'))
sign_image = Label(top)
def classify(file_path):
    # print(file_path)
    Str="Main1.py --input "+file_path+" --output outputVideos/bridgeOut.avi --yolo yolo-"
    # harActivity = "Main1.py --input inputVideos/bridge.mp4 --output outputVideos/bridgeOut.avi --yolo yolo-"
    subprocess.call("python "+Str)


def show_classify_button(file_path):
    classify_b=Button(top,text="Get RealTime Vechicle Reading",command=lambda: classify(file_path),padx=10,pady=5)
    classify_b.configure(background='#364156', foreground='white',font=('arial',10,'bold'))
    classify_b.place(relx=0.79,rely=0.46)
    button2_canvas = canvas1.create_window( 800, 300, anchor = "nw", window = classify_b)
```

```
def upload_video():
    try:
        file_path=filedialog.askopenfilename()
        label.configure(text='')
        show_classify_button(file_path)
    except:
        pass
upload=Button(top,text="UploadA
Video",command=upload_video,padx=10,pady=5)
upload.configure(background='#364156',
foreground='white',font=('arial',10,'bold'))

upload.pack(side=BOTTOM,pady=50)
button1_canvas = canvas1.create_window( 600, 500, anchor =
"nw", window = upload)

sign_image.pack(side=BOTTOM,expand=True)
label.pack(side=BOTTOM,expand=True)
heading = Label(top, text="RealTime Vechicle Project",pady=20,
font=('arial',20,'bold'))
heading.configure(background='#CDCDCD',foreground='#FF000
0')
heading.pack()
button2_canvas = canvas1.create_window( 500, 100, anchor =
"nw", window = heading)

top.mainloop()
```

CHAPTER 6
EXPERIMENTAL RESULTS

FIG:6.1 uploading video into the application

FIG:6.2 detecting of the vehicles



CHAPTER 7
CONCLUSION &FUTURE ENHANCEMENT

7.1 Conclusion

The successful implementation of video analysis for real-time vehicle classification and counting demonstrates the potential of computer vision and deep learning technologies in transforming traffic management systems. By utilizing real-time video feeds and advanced object detection models, our system accurately identifies and categorizes various vehicle types while maintaining high processing speed and reliability.

This project not only provides a scalable and cost-effective solution for intelligent traffic monitoring but also opens doors to further enhancements such as congestion prediction, adaptive signal control, and integration with smart city infrastructure. Future improvements may include the use of more sophisticated AI models, edge computing for faster response times, and expansion to multi-camera or drone-based surveillance.

Overall, this project highlights the potential of deep learning, specifically autoencoder-based architectures, in real-world image enhancement applications. Future work could focus on integrating attention mechanisms, exploring convolutional autoencoders for spatial feature learning, or adapting the model for real-time video denoising.

7.2 Future Enhancement

For future enhancements, several directions can be considered to further improve the age estimation system using deep learning models. Given the demonstrated benefits of fine-tuning and hyperparameter optimization, future work could explore more advanced or alternative optimization algorithms beyond PSO, such as Bayesian optimization or genetic algorithms, to potentially achieve better results with reduced execution time. Additionally, experimenting with newer and more efficient deep learning architectures, such as EfficientNet or Vision Transformers, may lead to further improvements in accuracy and computational efficiency. Expanding the dataset and employing sophisticated data augmentation techniques could also enhance the model's generalization and robustness. Moreover, implementing ensemble methods that combine the strengths of multiple models might help achieve more consistent and reliable age predictions. Finally, attention should be given to optimizing the trade-off between model performance and computational cost, ensuring that the chosen optimization methods are practical for real-world deployment. By pursuing these enhancements, the system can become more accurate, efficient, and applicable to a wider range of scenarios.

REFERENCES

[1] Brown, T., & White, M. (2019). "Deep Learning for Traffic Monitoring Systems." This study explores CNN architectures for vehicle detection.

[2] Singh, R., & Patel, K. (2020). "AI-Based Vehicle Classification in Smart Cities." The research discusses AI-driven vehicle categorization techniques.

[3] Nguyen, L., & Sharma, P. (2021). "YOLO-Based Vehicle Detection for Traffic Analytics." This paper evaluates YOLO models for real-time vehicle analysis.

[4] Martinez, D., & Johnson, S. (2022). "Intelligent Transportation Systems Using Computer Vision." The authors present AI-driven traffic management solutions.

[5] Garcia, H., & Lewis, F. (2023). "Real-Time Vehicle Counting and Road Traffic Analysis." This study examines AI techniques for improving traffic flow monitoring.

[6] Chen, W., & Roberts, L. (2023). "Edge Computing for AI-Powered Traffic Surveillance." The paper explores AI-driven geospatial data processing.